

Product Reviews Sentiment Analysis

"Decoding customer voices, discovering
insights beyond the stars."

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Our Agenda

- 1. Project Summary & Problem Statement**
- 2. Exploratory Data Analysis (EDA)**
- 3. Data Cleaning & Preprocessing**
- 4. Deep Learning Models & Evaluation**
- 5. Conclusion, Recommendations & Future Work**



Project Summary & Problem Statement

Business Problem

Imagine an Egyptian food e-commerce company in a fast-growing market. Thousands of daily customer reviews reveal what delights or disappoints them. This project turns raw reviews into actionable insights, helping the company improve products, marketing strategies, and customer satisfaction—boosting both the company's performance and Egypt's e-commerce ecosystem.



Data Overview



Dataset : Amazon Fine Food Reviews

About this Dataset : The dataset contains Amazon food product reviews with ratings, summaries, and full customer comments. It is widely used for sentiment analysis and text-classification research, helping model real customer opinions and evaluate NLP models on large, real-world review data.

Data Shape: (568454, 10)

Columns Overview

Column	Description
Id	unique identifier for each review
ProductId	Unique identifier for the product being reviewed
UserId	Unique identifier for the user who wrote the review
ProfileName	Name of the reviewer (profile name)
HelpfulnessNumerator	Number of users who found this review helpful
HelpfulnessDenominator	Number of users who indicated whether they found the review helpful or not
Score	Rating given by the user (from 1 to 5)
Time	UNIX timestamp when the review was posted
Summary	Short summary of the review
Text	Full text of the review



Exploratory Data Analysis (EDA)

Q1: How many products, users and reviews are there in the company?

Total Products

74257

Total Users

256042

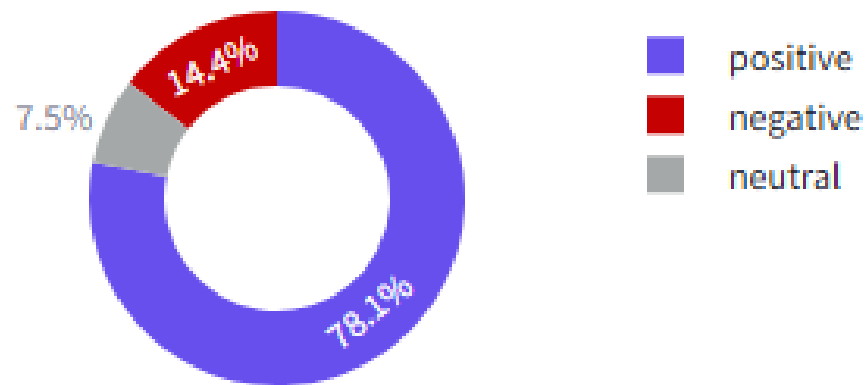
Total Reviews

568401

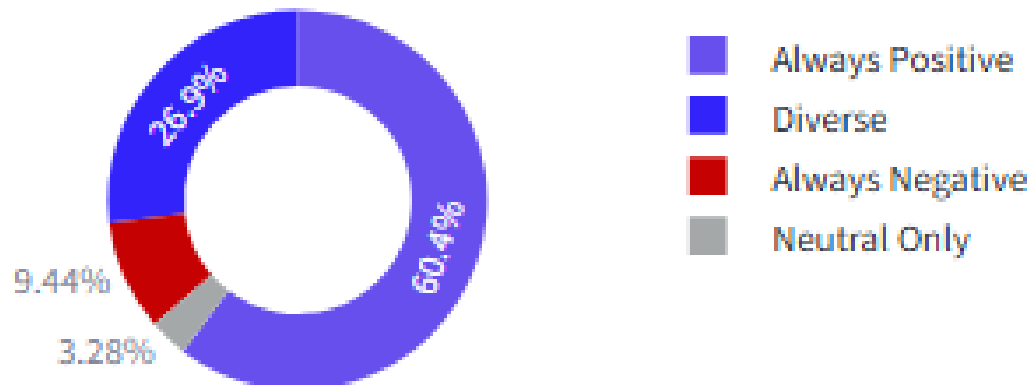
Q2: What is the distribution of Customer Sentiment and Behavior in the data?



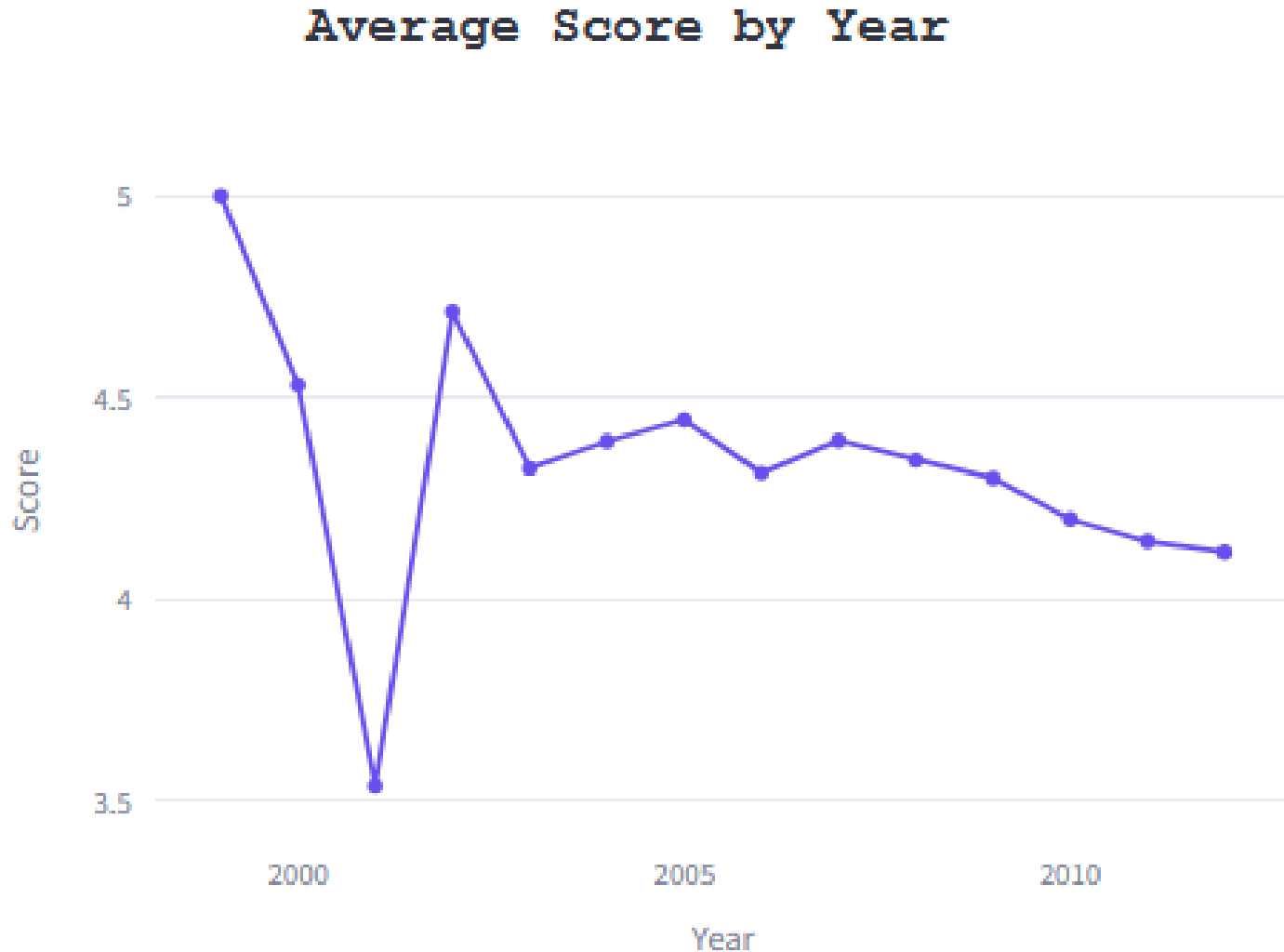
Customer Sentiment



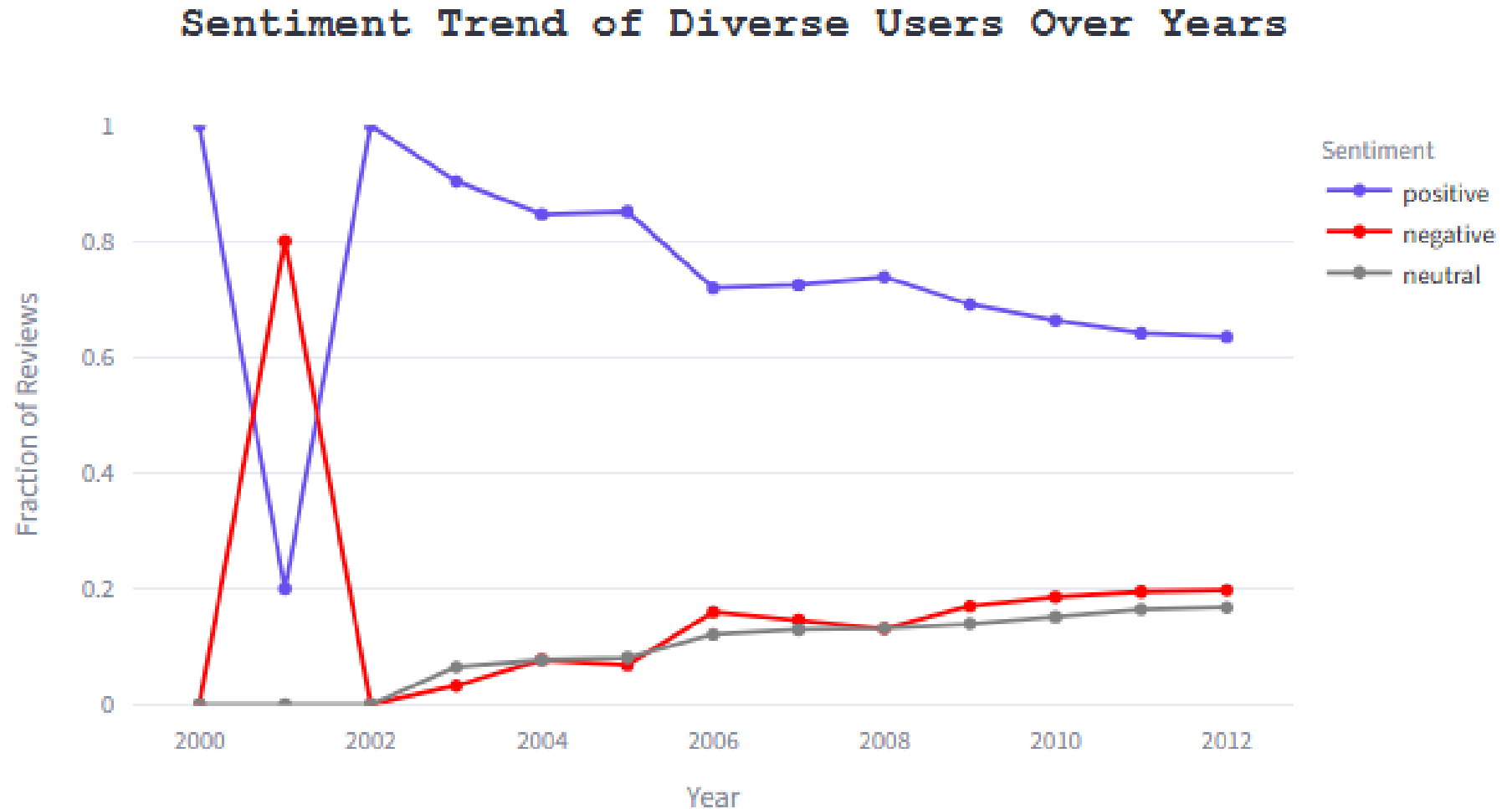
Customer Behavior



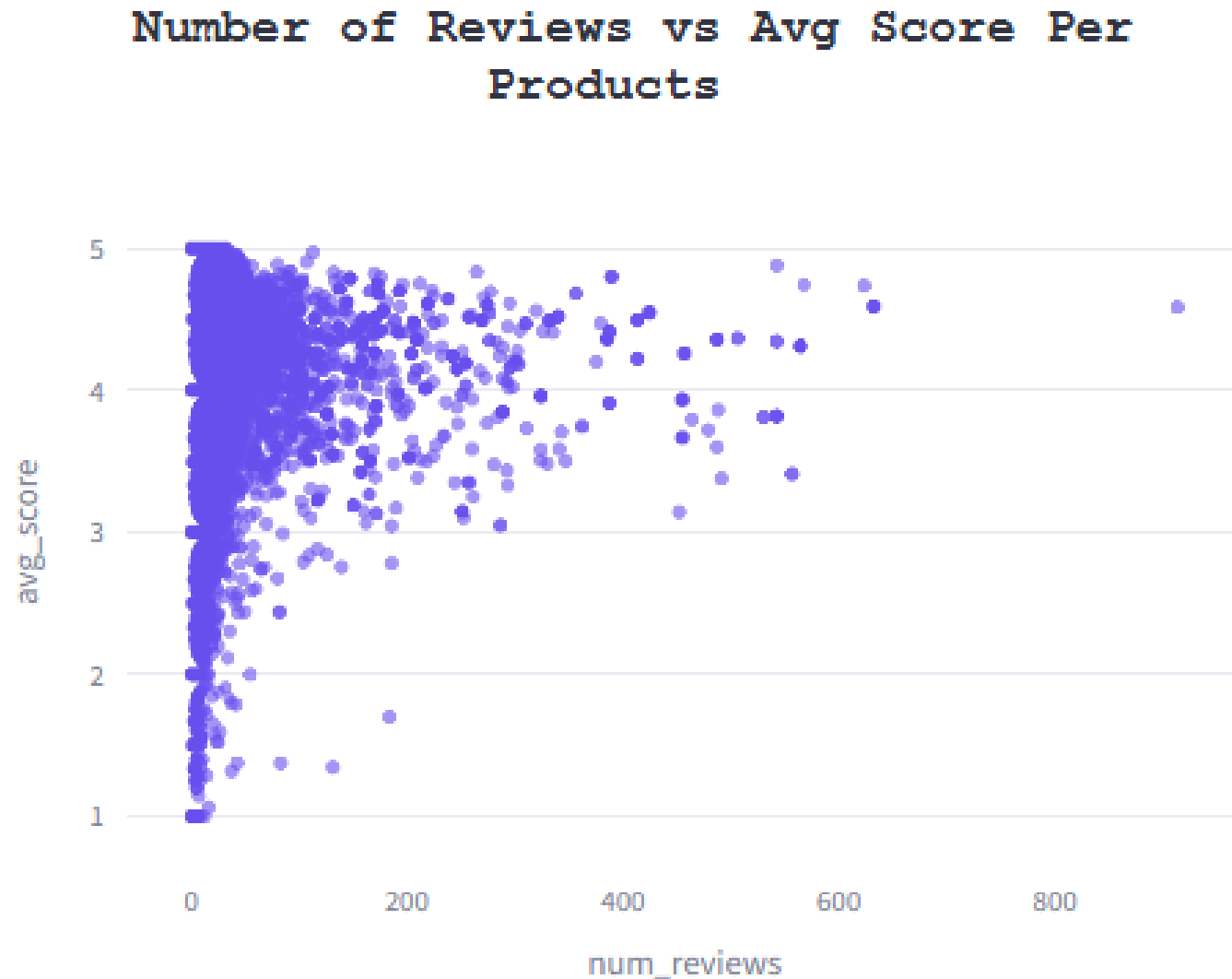
Q3: How has the average product score evolved over the years?



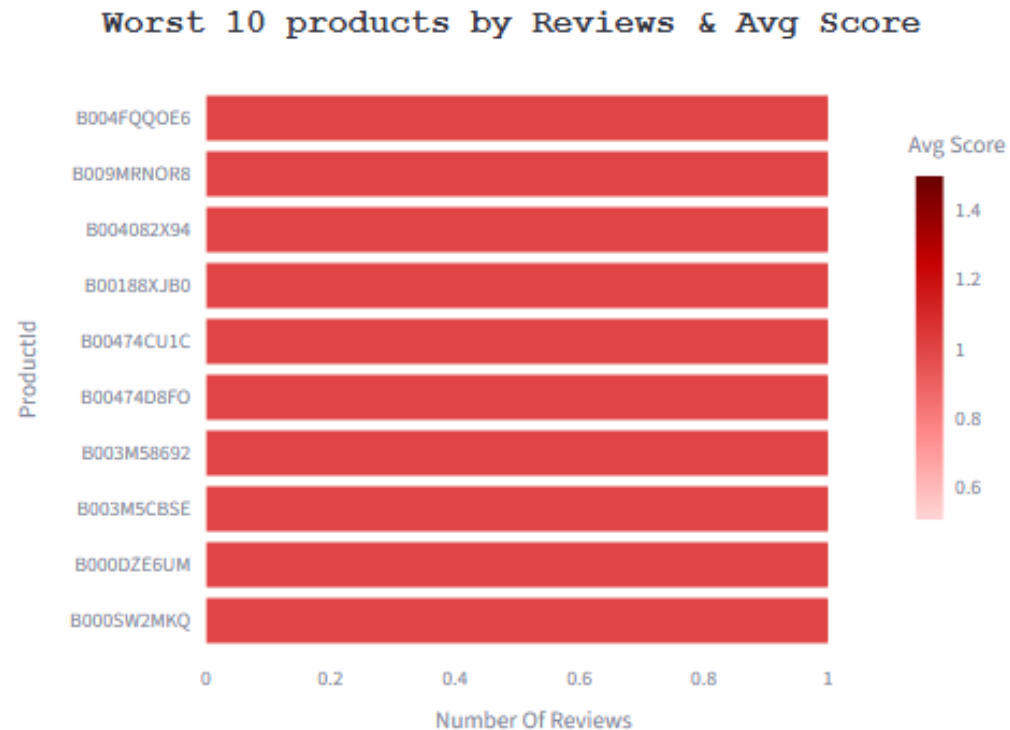
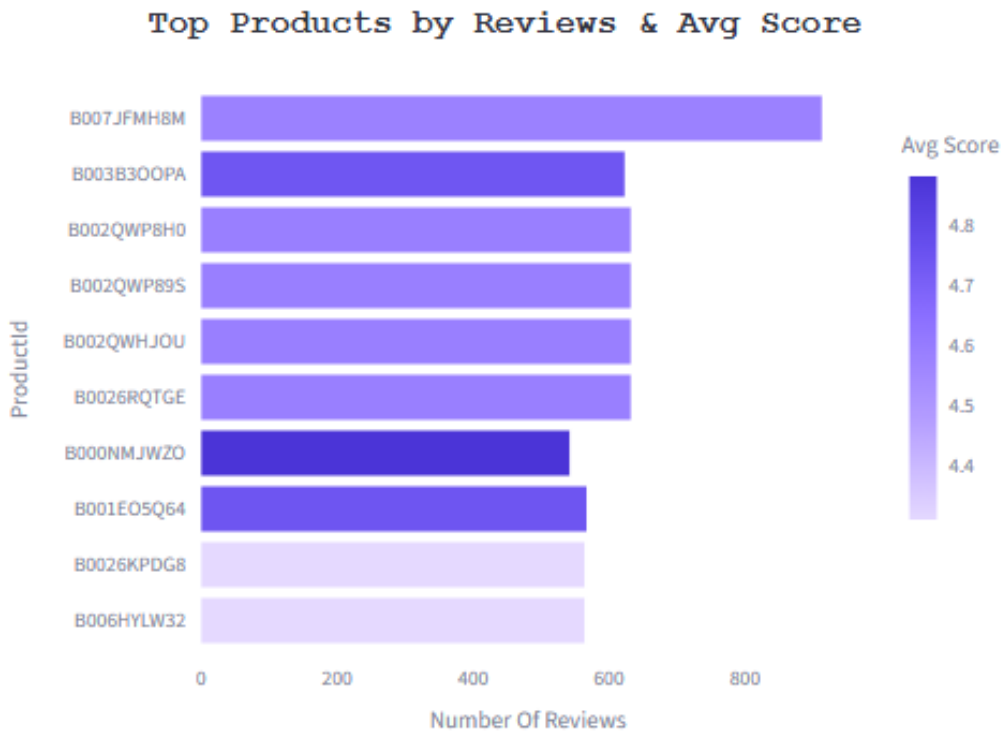
Q4: How has the sentiment of diverse users changed over the years?



Q5: Is there a correlation between the number of reviews and average score for each product?



Q6: Which products are performing best and worst based on number of reviews and average scores?



Data Cleaning & Preprocessing

1] Feature Engineering



- Created sentiment feature

Sentiment = positive / neutral / negative based on Score

- Created Year feature

Year = represents the year in which the review was written

- Created behavior feature

Behavior = customer behavior category based on sentiment history (Always Positive, Always Negative, Neutral Only, Diverse)

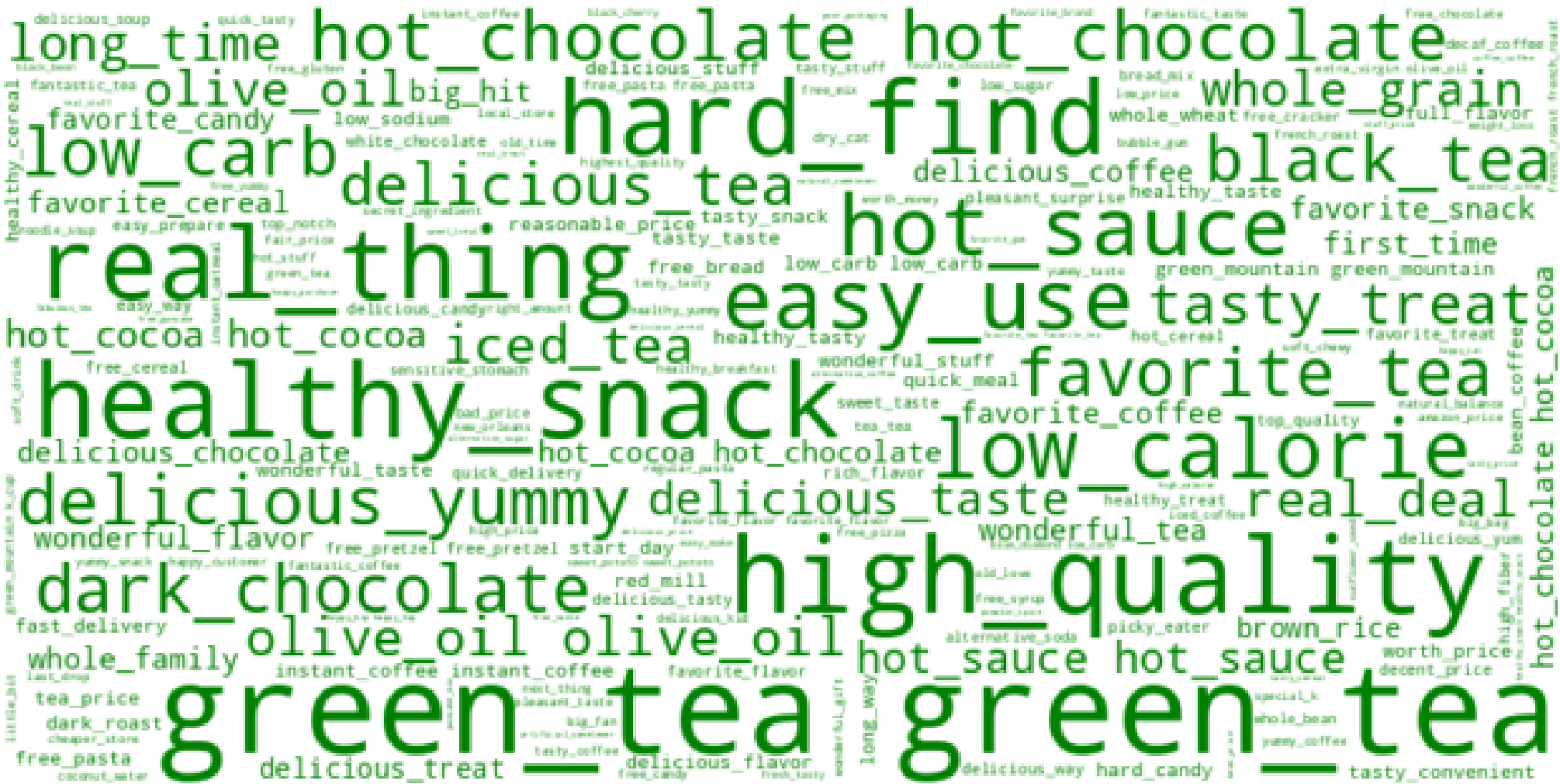
2] Text Processing :

- Performed text cleaning by removing **URLs, HTML tags, emojis, non-English characters, and extraneous whitespace.**
- Applied **BERT** WordPiece tokenization to effectively represent unknown, rare, misspelled, and morphologically complex words.
- Utilized **BERT's** contextual tokenizer, enabling self-attention mechanisms that capture word relationships and subtle contextual meaning.
- Implemented token padding to maintain fixed sequence lengths and optimize computational efficiency during training.
- Employed attention masks to ensure the model attends only to valid tokens while ignoring padded positions.

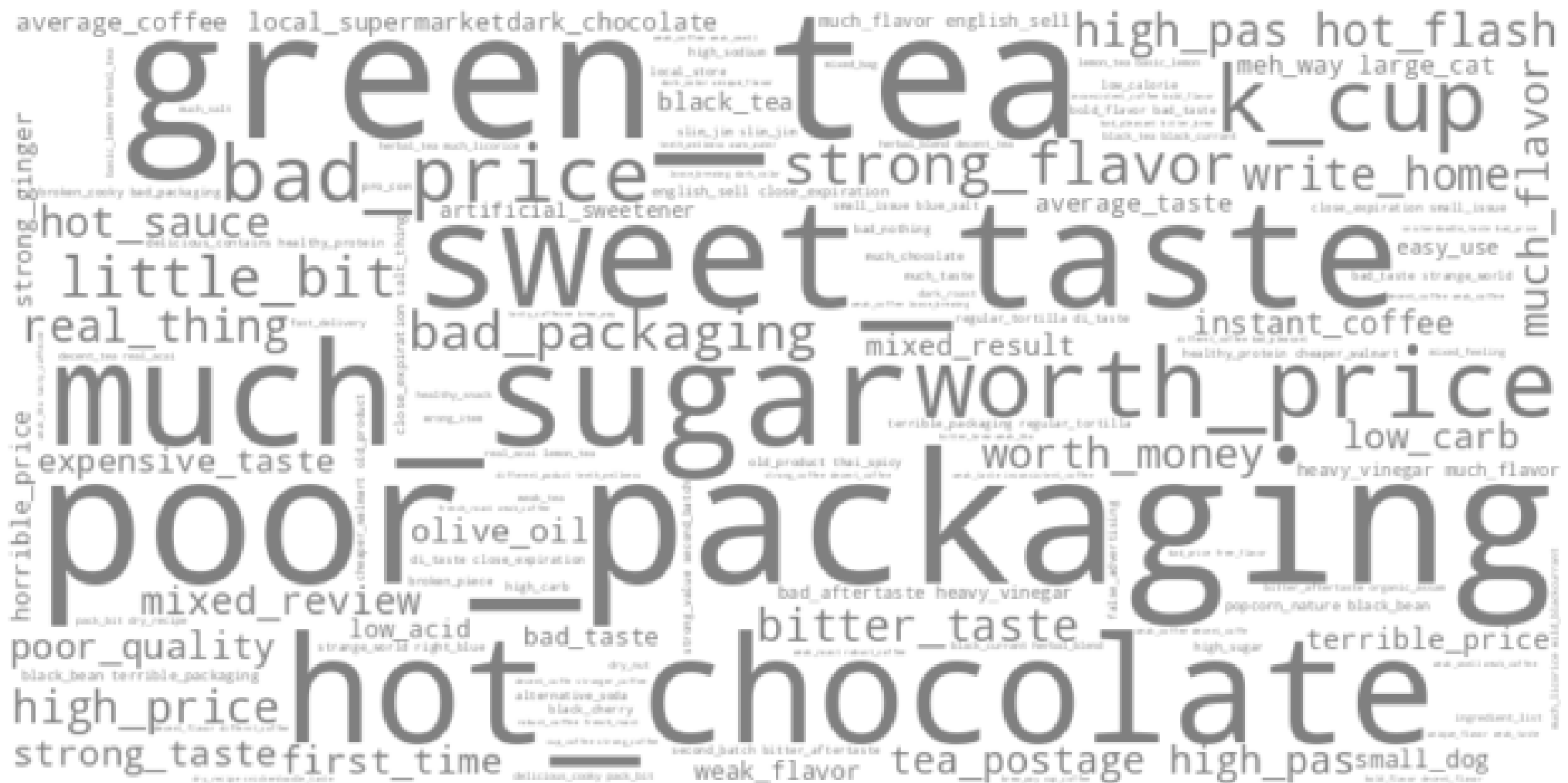
Word cloud for negative sentiment



Word cloud for positive sentiment



Word cloud for neutral sentiment



4] Handling Class Imbalance

- Used algorithm-level **class weighting**.
- Safe single-pass oversampling.
- Added optional **focal loss**.



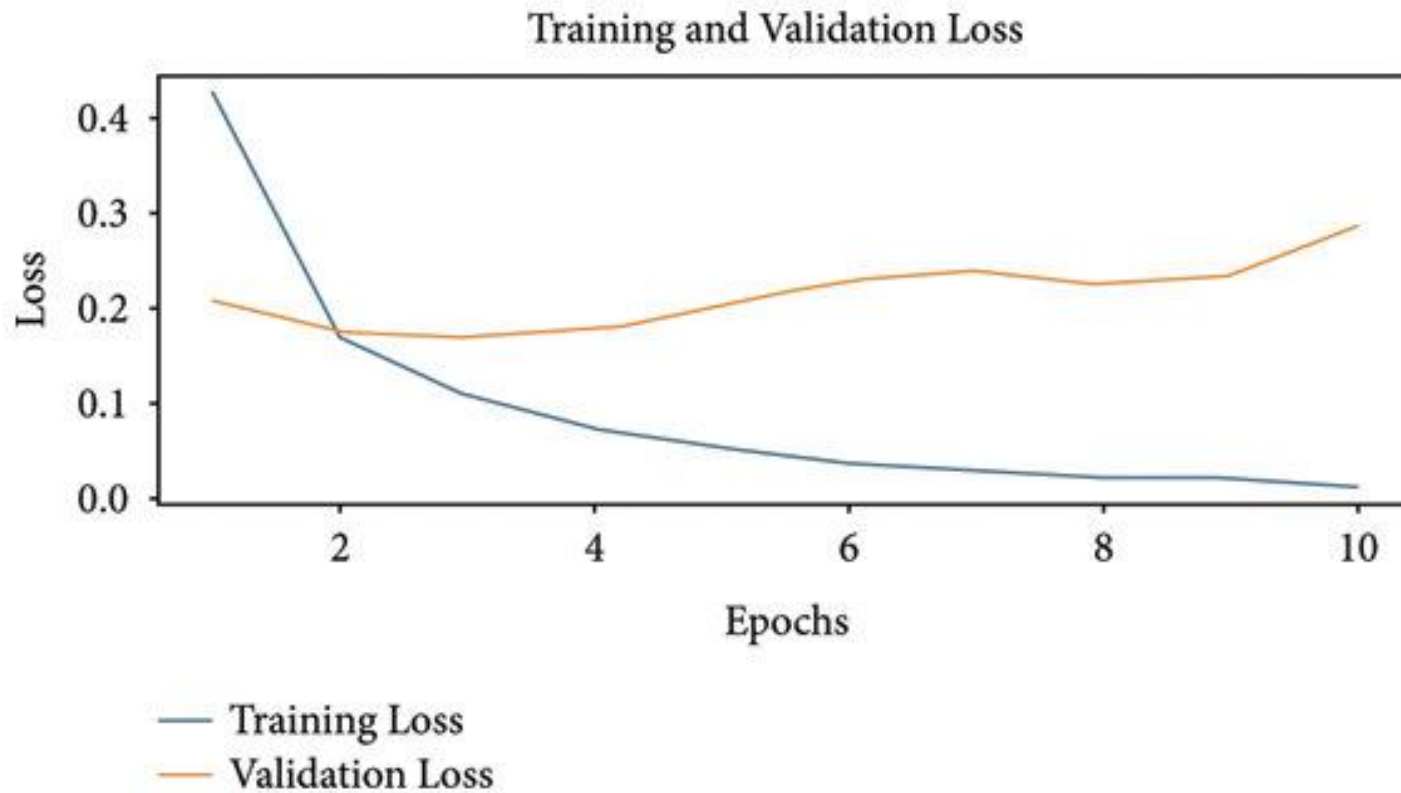


Modeling & Evaluation

1] Baseline: BERT

- Used **BERT** to verify pipeline
- Used as **Full Transformer**
- Used as **Tokenizer** and **Classifier**
- Got **Accuracy = 0.9** and **F1_Score = 0.77**
- Revealed imbalance problem → motivated improvement

1] Baseline: BERT



Mild Overfitting

2] Model Upgrade: RoBERTa

Why RoBERTa?

- **Dynamic masking** → better contextual learning
- **Larger batch + longer training** → stronger language representations
- **Trained on more data** → more robust model
- **No next-sentence prediction** → more efficient training
- **Better tokenizer (BPE)** → handles vocabulary more flexibly

2] Model Upgrade: RoBERTa

Hyperparameter Searching

- Ran grid search on 5% dataset
- Searched: learning rate, batch, accumulation, weight decay and epochs
- Best configuration:
 - LR = 5e-5, Batch = 4
 - Accum = 8
 - Epochs = 3
 - Weight Decay = 0.05

2] Model Upgrade: RoBERTa

More Unbalanced Data handling:

Solution 1: Controlled Oversampling

Replicated minority-class samples **carefully**, with a safety limit (`max_expand_ratio`) to avoid exploding dataset size.

Solution 2: Focal Loss

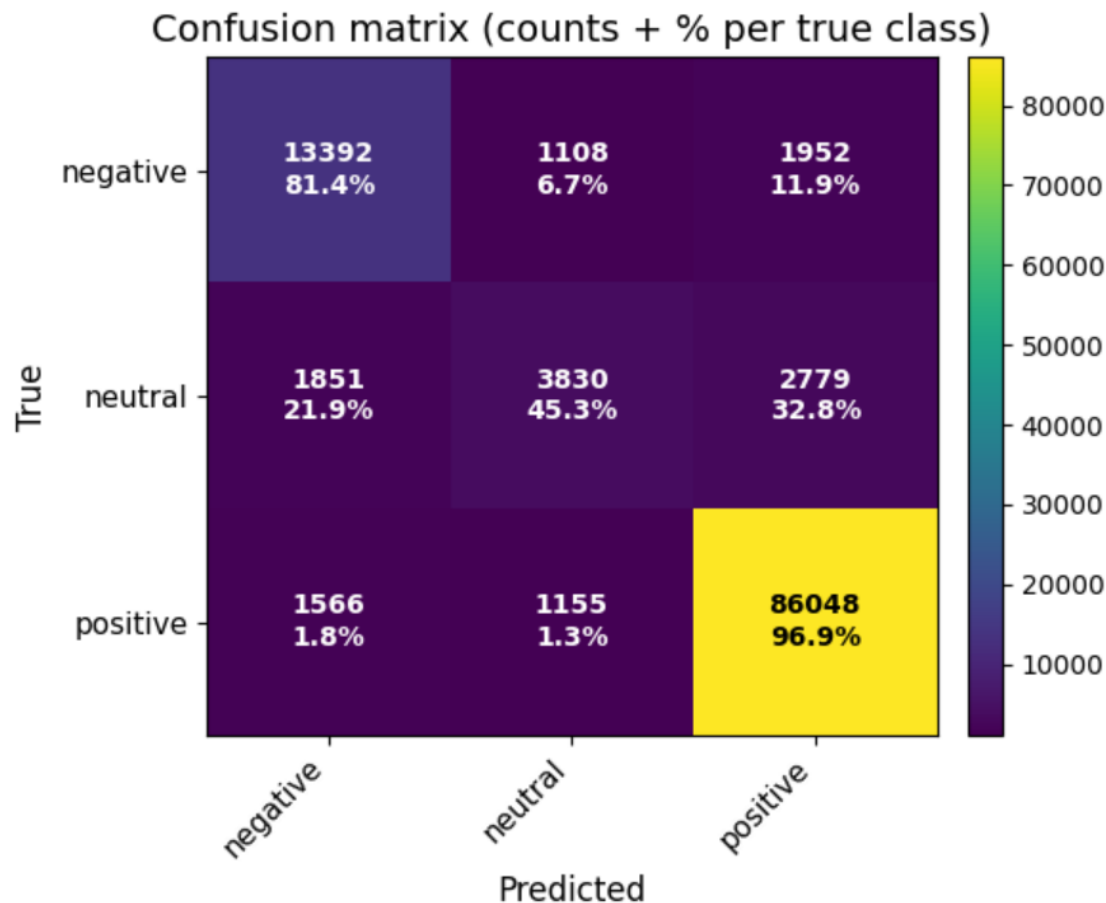
Modified the loss function to **down-weight easy examples** and focus more on **hard/minority-class mistakes**.

2] Model Upgrade: RoBERTa

The exploration of more advanced imbalance-mitigation methods was constrained by available resources, as current training requires more than 30 hours.

2] Model Upgrade: RoBERTa

Results





Recommendations & Future Work

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1. Expand the Classification Task (3 → 5 Classes)

- Move from **sentiment polarity** (positive / neutral / negative) to **fine-grained star rating prediction** (1 ★ to 5 ★).
- This enables more detailed understanding of customer satisfaction.

2. Try using a pretrained model to generate more rows for neutral data to solve unbalance

3. Try larger models (RoBERTa-large, DeBERTa).

4. Explore the VADER Transformer

Recommendations & Future Work

Before: **VADER** (Valence Aware Dictionary and sEntiment Reasoner) is a rule-based sentiment tool widely used for social media and short texts.

VADER Transformer refers to transformer models fine-tuned to mimic or improve upon VADER's lexicon-based style — useful for short, informal, or emoji-rich text.

Plan: compare **RoBERTa**, **BERT**, and a **VADER-inspired transformer** on the same dataset for deeper benchmarking.

Dashboard QR Code



StreamLite Deployment QR Code



Thank you!

